

Problem Statement and Goals

ProgName

Team #, Team Name
Student 1 name
Student 2 name
Student 3 name
Student 4 name

Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

1 Problem Statement

[You should check your problem statement with the [problem statement checklist](#) —SS].

[You can change the section headings, as long as you include the required information. —SS]

1.1 Problem

[What problem are you trying to solve? Why is it important? Do not talk about how you will solve the problem, only what the problem is. AI is rarely part of the problem. The problem is “what” you want to do, not “how” you want to do it. The problem statement should be written in a way that is understandable to a non-technical audience. —SS]

1.2 Inputs and Outputs

[Characterize the problem in terms of “high level” inputs and outputs. Use abstraction so that you can avoid details. —SS]

1.3 Stakeholders

1.4 Environment

1.4.1 Hardware Environment

The simulator will run on a compatible VR headset capable of supporting real-time 3D applications. The headset will provide the display and motion tracking required for the user to interact with the simulated environment. VR controllers or compatible FPV flight controls will be used to control the simulated drone.

1.4.2 Software Environment

The simulator will run on the VR headset using a VR-compatible operating system and runtime environment. The software will require the necessary VR support to provide real-time graphics, motion tracking, and user input. The simulator will operate as a standalone VR application on supported hardware.

2 Goals

1. **Provide a Realistic FPV Drone Flight Experience:** Create an immersive virtual reality FPV drone simulation that maintains a smooth and consistent frame rate during normal gameplay.
2. **Support Beginner and Advanced Flight Training:** Enable new users to develop fundamental FPV flight skills while allowing experienced pilots to refine their performance through improvements in course completion times, flight consistency, and crash frequency.
3. **Provide Realistic and Responsive Flight Controls:** Deliver intuitive controls with minimal input delay and realistic, consistent drone responses to throttle, pitch, roll, and yaw inputs.
4. **Provide Varied Flight Environments and Courses:** Offer multiple environments and courses featuring different layouts, obstacles, and flight challenges to provide varied training scenarios.
5. **Track Flight Performance:** Record basic flight performance data, including completion times, crashes, and course completion status, allowing users to review and monitor their progress.

3 Stretch Goals

1. **Provide Multiple Drone Configurations:** Offer multiple drone configurations with distinct flight characteristics, handling, and responsiveness, allowing users to experience different drone setups and adapt their flying techniques accordingly.

2. **Support Multiplayer Flight:** Enable multiple users to operate drones within the same virtual environment, allowing them to practice and interact together in shared training scenarios.
3. **Provide AI-Based Flight Analysis and Feedback:** Analyze controller inputs and flight performance to identify patterns such as inefficient inputs, inconsistent control, excessive corrections, and other areas for improvement, and provide personalized feedback to help users refine their flying skills.

4 Custom Documents (CDs)

The extras chosen are the Design Thinking Report and the Usability Report. The Design Thinking Report will explain the intent behind our high level decisions and direction behind our design. The usability report will show the results of usability testing while also showing the response received from users. Our goals and methods regarding the testing will be documented as well.

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?